

DSAA3073: Theories in Data Science

Week 2: PAC Learning Framework & VC Dimension

Concise Learning Sheet

Wei Wang · DSA, HKUST(GZ) · 2026-06-27

Explorative projection

Opening Challenge

When Can We Trust a Learned Model?

You train a neural network on 1000 images and achieve 99% training accuracy. Should you deploy it in production?

- What guarantees can you make about test performance?
- If you had 10,000 images instead, how would your confidence change?
- Does the answer depend on whether you used 10 neurons or 10,000 neurons?
- Can you quantify the trade-off between model complexity and sample size?

This week formalizes these questions through the PAC learning framework and introduces VC dimension as the fundamental measure of hypothesis class complexity.

Introduction

Last week we built the probabilistic toolkit: concentration inequalities that bound how sample averages deviate from expectations. This week we apply these tools to answer the central question of learning theory: **When is learning possible?**

← From Week 1: Concentration Inequalities

Last week's concentration inequalities told us that sample averages converge to expectations. But how many samples? And what if we're choosing between millions of hypotheses? This week answers: learning is possible when hypothesis class complexity—measured by VC dimension—is finite, no matter how large the class cardinality.

Learning objectives: By the end of this week, you will:

1. Formalize learning problems using the PAC framework
2. Derive sample complexity bounds for finite hypothesis sets
3. Compute VC dimension for geometric concept classes
4. Apply Sauer's lemma to bound growth functions
5. Understand the fundamental connection between VC dimension and generalization

The PAC Learning Framework

Before proving anything is learnable, we need to define what “learnable” means. The PAC framework formalizes three requirements: (1) polynomial sample complexity, (2) low generalization error with high probability, (3) computational efficiency. We focus on (1) and (2) this week.

Setup and Notation

Definition: Learning Problem Components

A binary classification learning problem consists of:

- **Input space** $\mathcal{X}$: the set of all possible examples (e.g., images, text documents, sensor readings)
- **Label space** $\mathcal{Y} = \{0, 1\}$: binary labels (e.g., spam/not spam, disease/healthy)
- **Concept** $c : \mathcal{X} \rightarrow \mathcal{Y}$: the true labeling function that assigns the correct label to each input (unknown to the learner)
- **Concept class** $\mathcal{C}$: the set of possible target concepts we wish to learn (the “ground truth” can be any function in this set)
- **Hypothesis set** $\mathcal{H}$: the set of functions the learner can choose from (our model class)
- **Distribution** $\mathcal{D}$: a fixed but unknown probability distribution over the input space $\mathcal{X}$ (determines which examples are likely)
- **Sample** $S = \{(x_1, y_1), \dots, (x_m, y_m)\}$: a training dataset of m labeled examples drawn i.i.d. (**independent and identically distributed**), where each $x_i \sim \mathcal{D}$ and $y_i = c(x_i)$

Typical case: Often $\mathcal{H} = \mathcal{C}$, meaning we search for the target concept within the same class it comes from.

Notation Map

Symbol	Meaning	Distinguish from
$\mathcal{X}$	Input/domain space	$\mathcal{Y}$: label space, usually $\{0, 1\}$ here
c	True target concept	h : learner’s chosen hypothesis
$\mathcal{C}$	Concept class: possible target concepts	$\mathcal{H}$: hypothesis set searched by the learner
$\mathcal{D}$	Unknown data distribution over inputs	S : the finite training sample drawn from $\mathcal{D}^m$
$R(h)$	Generalization error under $\mathcal{D}$	$\hat{R}_S(h)$: empirical error on S
$\mathbb{P}$	Probability over the currently implied randomness	$\mathbb{P}_S$: probability over the random draw of training sample S
ε	Allowed error tolerance	δ : allowed failure probability
Realizable	Some $h^* \in \mathcal{H}$ has $R(h^*) = 0$	Agnostic: no perfect hypothesis is assumed

Example: Disease Diagnosis from Blood Test

- $\mathcal{X} = \mathbb{R}^+$: biomarker concentration levels
- c : true disease/healthy threshold (unknown to us)
- $\mathcal{H} = \{h_t : h_t(x) = \mathbb{1}_{x > t}, t \in \mathbb{R}\}$: threshold classifiers
- $\mathcal{D}$: distribution of biomarker levels in population
- S : 100 labeled patient samples

Key question: How many patients must we test before trusting our learned threshold?

Error Measures

Before defining when learning succeeds, we need to measure error. We distinguish between **generalization error** (true error on the population) and **empirical error** (observed error on the training sample).

Definition: Generalization Error

The **generalization error** (also called **true risk** or **test error**) of hypothesis h is:

$$R(h) = \mathbb{P}_{x \sim \mathcal{D}}[h(x) \neq c(x)] = \mathbb{E}_{x \sim \mathcal{D}}[\mathbb{1}_{h(x) \neq c(x)}] \quad (1)$$

In words: The probability that h makes a mistake on a randomly drawn example from the true distribution $\mathcal{D}$. This is what we truly care about but cannot directly compute (since we don't know $\mathcal{D}$).

Definition: Empirical Error

The **empirical error** (also called **training error**) of h on sample S is:

$$\hat{R}_S(h) = \frac{1}{m} \sum_{i=1}^m \mathbb{1}_{h(x_i) \neq y_i} \quad (2)$$

In words: The fraction of training examples that h misclassifies. This is computable from the sample but only approximates $R(h)$.

Key relationship: $\hat{R}_S(h)$ is an unbiased estimator of $R(h)$ for a fixed hypothesis h (chosen before seeing the data): $\mathbb{E}_S[\hat{R}_S(h)] = R(h)$. However, when h is chosen based on S (as in learning), this unbiasedness no longer directly applies—this is why we need more sophisticated analysis.

Pause and Think

Question: Why is $\hat{R}_S(h)$ an unbiased estimator of $R(h)$ for a fixed hypothesis h chosen before seeing the sample?

Hint: Each indicator is a Bernoulli random variable. Use linearity of expectation. If h is chosen after seeing S , this unbiasedness argument no longer applies directly.

Answer:

$$\mathbb{E}[\hat{R}_S(h)] = \mathbb{E}\left[\frac{1}{m} \sum_{i=1}^m \mathbb{1}_{h(x_i)=y_i}\right] \stackrel{\text{By linearity}}{=} \frac{1}{m} \sum_{i=1}^m \mathbb{E}[\mathbb{1}_{h(x_i)=y_i}] = \frac{1}{m} \sum_{i=1}^m \mathbb{P}(h(x_i)=y_i) = \mathbb{P}(h(x)=y) = R(h)$$

PAC Learnability

PAC stands for **Probably Approximately Correct**—a hypothesis that is approximately correct (error $\leq \varepsilon$) with high probability (at least $1 - \delta$).

Definition: PAC Learnability (Realizable Case)

A hypothesis class $\mathcal{H}$ is **PAC-learnable in the realizable setting** if there exists an algorithm A and a **sample-complexity function** $m_{\mathcal{H}}(\varepsilon, \delta)$ (polynomial in $\frac{1}{\varepsilon}$ and $\frac{1}{\delta}$ for efficient classes) such that:

For all $\varepsilon > 0$, $\delta > 0$, all distributions $\mathcal{D}$ over $\mathcal{X}$, and all target concepts $c \in \mathcal{H}$:

If $S = \{(x_i, c(x_i))\}_{i=1}^m$ with $x_i \sim \mathcal{D}$ i.i.d. and $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$, then with probability at least $1 - \delta$ over $S \sim \mathcal{D}^m$:

$$R(A(S)) \leq \varepsilon \tag{3}$$

In words: Given enough samples ($m \geq m_{\mathcal{H}}(\varepsilon, \delta)$), the algorithm A outputs a hypothesis with generalization error at most ε , and this holds with probability at least $1 - \delta$ over the random training sample.

Realizable assumption: We assume there exists a perfect hypothesis $h^* \in \mathcal{H}$ with $R(h^*) = 0$ (the target concept belongs to our hypothesis class). The agnostic case (Week 3) removes this assumption and only requires competing with the best hypothesis in $\mathcal{H}$.

Interpretation of PAC:

- **Probably** ($1 - \delta$): the guarantee holds with high confidence over random training samples
- **Approximately** (ε): we allow small error tolerance (not necessarily zero error)
- **Correct:** low generalization error on unseen data

Mental Model: PAC Learning Taxonomy

Three dimensions of learning problems:

Dimension	Choices	Impact
Realizability	Realizable vs Agnostic	$O(\frac{1}{\epsilon})$ vs $O(\frac{1}{\epsilon^2})$
Complexity	Finite $ \mathcal{H} $ vs Infinite	$\log \mathcal{H} $ vs $d \log(m)$
Algorithm	Consistent vs ERM	Works when realizable vs always

Decision tree for choosing the right bound:

Is hypothesis class finite or infinite?

```

├ Finite |H|
│   ├── Is problem realizable (perfect h* exists)?
│   │   ├── Yes → Use realizable bound:  $O(\log|H|/\epsilon)$ 
│   │   └── No → Use agnostic bound:  $O(\log|H|/\epsilon^2)$ 
│   └── Algorithm: Consistent learner (realizable) or ERM (agnostic)
└ Infinite |H|
    ├── Is VC dimension finite?
    │   ├── Yes → Use VC bound:  $O(d/\epsilon^2)$ 
    │   └── No → NOT PAC-learnable!
    └── Algorithm: ERM over hypothesis class
  
```

Backbone message: From counting hypotheses to counting behaviors—we care about effective complexity (VC dimension), not cardinality.

Connection

Key insight: PAC learning requires polynomial sample complexity in $\frac{1}{\epsilon}$ and $\frac{1}{\delta}$. Exponential dependence would be impractical.

Progress on backbone: The PAC framework formalizes our opening challenge: “What can you guarantee about performance on new data?” Answer: with $m \geq m_{\mathcal{H}}(\epsilon, \delta)$ samples, we guarantee error $\leq \epsilon$ with probability $\geq 1 - \delta$. This transforms machine learning from empirical art to rigorous science. The central question becomes: **what is $m_{\mathcal{H}}(\epsilon, \delta)$?** We’ll see it depends on hypothesis class complexity.

We’ve defined PAC learning, but haven’t answered **when** it’s achievable. We start with the simplest case: finite hypothesis sets with perfect hypotheses.

Finite Hypothesis Sets: Realizable Case

Suppose we're searching through a large but finite list of candidate models. In the realizable setting, we know one of them is perfect ($R(h^*) = 0$). The question: how many samples before we can trust any perfect-looking model?

The Consistent Learner

Definition: Realizable Setting

We say the problem is **realizable** if there exists $h^* \in \mathcal{H}$ with $R(h^*) = 0$. That is, the target concept c is in the hypothesis set.

Algorithm: Return any $h_S \in \mathcal{H}$ with $\hat{R}_S(h_S) = 0$ (consistent with training data). In the realizable setting, at least one such hypothesis exists because the target concept is in $\mathcal{H}$.

Theorem: PAC Bound for Finite $\mathcal{H}$ (Realizable)

Let $\mathcal{H}$ be finite in the realizable setting. For any $\varepsilon, \delta > 0$, any distribution $\mathcal{D}$ over $\mathcal{X}$, and any target concept $c \in \mathcal{H}$:

If $m \geq \frac{1}{\varepsilon} (\log |\mathcal{H}| + \log(\frac{1}{\delta}))$, then with probability at least $1 - \delta$ over $S \sim \mathcal{D}^m$, every hypothesis consistent with S has true error at most ε :

$$\forall h \in \mathcal{H} : \hat{R}_S(h) = 0 \implies R(h) \leq \varepsilon \quad (4)$$

Thus any algorithm that returns a consistent h_S also satisfies $R(h_S) \leq \varepsilon$ with probability at least $1 - \delta$.

Proof (Guided Discovery):

Step 1: Define bad hypotheses.

Let $\mathcal{H}_{\text{bad}} = \{h \in \mathcal{H} : R(h) > \varepsilon \text{ and } \hat{R}_S(h) = 0\}$. These are hypotheses that fool us—they look perfect on training data but fail on new data.

Step 2: Bound probability that one bad hypothesis is consistent.

Fix $h \in \mathcal{H}_{\text{bad}}$. What is $\mathbb{P}_S[\hat{R}_S(h) = 0]$?

Pause and Think

Question: If $R(h) > \varepsilon$, what's the probability that h makes no errors on m i.i.d. samples?

Hint: Each sample has probability $1 - R(h)$ of being classified correctly. Use independence.

Answer:

$$(5) \quad \mathbb{P}_S(\hat{R}_S(h) = 0) = \mathbb{P}[\text{all } m \text{ samples are correct}] = (1 - R(h))^m$$

Since $R(h) > \varepsilon$, we have:

$$\mathbb{P}_S[\hat{R}_S(h) = 0] = (1 - R(h))^m < (1 - \varepsilon)^m \quad (6)$$

Using the inequality $(1 - \varepsilon) \leq e^{-\varepsilon}$:

$$\mathbb{P}_S[\hat{R}_S(h) = 0] \leq e^{-\varepsilon m} \quad (7)$$

Step 3: Union bound over all bad hypotheses.

$$\begin{aligned} \mathbb{P}_S[\exists h \in \mathcal{H}_{\text{bad}} : \hat{R}_S(h) = 0] &\leq \sum_{h \in \mathcal{H}_{\text{bad}}} \mathbb{P}_S[\hat{R}_S(h) = 0] \\ &\leq |\mathcal{H}| \cdot e^{-\varepsilon m} \end{aligned} \quad (8)$$

Step 4: Solve for sample complexity.

We want this probability to be at most δ :

$$|\mathcal{H}| \cdot e^{-\varepsilon m} \leq \delta \quad (9)$$

Taking logarithms:

$$\log|\mathcal{H}| - \varepsilon m \leq \log \delta \quad (10)$$

$$m \geq \frac{1}{\varepsilon} \left(\log|\mathcal{H}| + \log\left(\frac{1}{\delta}\right) \right) \quad (11)$$

Key Takeaway

Sample complexity scales **logarithmically** with $|\mathcal{H}|$. Doubling the hypothesis set size only requires a constant number of additional samples. The key term is $\log|\mathcal{H}|$, which measures the “effective complexity” of the hypothesis class.

Proof Template: Union Bound Pattern

This proof exemplifies a standard strategy we’ll use repeatedly:

1. Define “bad” set (objects that would violate our goal)
2. Bound probability for one bad object
3. Union bound over all bad objects
4. Solve for sample size

Examples in this course: Finite H realizable (this section), Finite H agnostic (next section)

Student Challenge

Exercise: If $|\mathcal{H}| = 2^{100}$ (huge!), how many samples do we need for $\varepsilon = 0.1$ and $\delta = 0.05$? Compare to the naive bound $|\mathcal{H}|$.

Hint: Testing 2^{100} possible diagnostic rules requires only ≈ 100 samples—far fewer than the naive “test each rule” approach.

✓ **Checkpoint**

Before moving on: In one sentence, explain why we can learn from a finite hypothesis class even when $|\mathcal{H}|$ is astronomically large.

Why Logarithmic Dependence?

Wrong Path: Naive Approach

“We need to test each hypothesis, so we need $|\mathcal{H}|$ samples.”

Why this fails: We’re not testing hypotheses independently. The sample provides information about all hypotheses simultaneously through their predictions.

Correct intuition: Each sample eliminates roughly half of the bad hypotheses (those that misclassify it). After $\log|\mathcal{H}|$ samples, we’ve eliminated most bad hypotheses with high probability.

🔧 Construct Your Own

Build a finite hypothesis class example:

Design a diagnostic test problem:

- Input: 3 biomarkers (glucose, cholesterol, blood pressure)
- Hypothesis class: All threshold combinations
 - $\mathcal{H} = \{h_{t_1, t_2, t_3} : \text{predict disease if } \geq k \text{ biomarkers exceed thresholds}\}$
 - How many hypotheses? (Hint: discretize thresholds, choose $k \in \{1, 2, 3\}$)

Compute sample complexity:

1. If we discretize each biomarker into 10 levels, $|\mathcal{H}| = ?$
2. For $\varepsilon = 0.05$, $\delta = 0.01$, how many patients needed?
3. Compare realizable vs agnostic bounds

Extension: What if we allow conjunctions (“glucose $> t_1$ AND cholesterol $> t_2$ ”)?

Progress on backbone: In the realizable case, we proved that ERM (Empirical Risk Minimization) finds a near-optimal hypothesis with polynomial sample complexity. But the realizable setting assumed a perfect hypothesis exists. Real problems are messier: labels have noise, models are approximate. Can we still make guarantees when perfection is impossible?

Finite Hypothesis Sets: Agnostic Case

In spam filtering, no linear classifier achieves zero error—language is too complex, labels are noisy. The agnostic setting asks: can we still learn the **best possible** hypothesis in our class, even if it's imperfect?

The Agnostic Setting

Definition: Agnostic Learning

In the **agnostic** setting, we make no assumption about realizability. The best hypothesis in $\mathcal{H}$ may have $R(h^*) > 0$.

Goal: Find h_S such that $R(h_S) \approx \min_{h \in \mathcal{H}} R(h)$.

Example: linear classifiers are realizable if the true boundary is linear and labels are noiseless; they are agnostic if the labels are noisy or the true boundary is nonlinear.

Algorithm: Empirical Risk Minimization (ERM)

$$h_S = \operatorname{argmin}_{h \in \mathcal{H}} \hat{R}_S(h) \quad (12)$$

Theorem: PAC Bound for Finite $\mathcal{H}$ (Agnostic)

Let $\mathcal{H}$ be finite. For any $\varepsilon, \delta > 0$ and any distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$:

If $m \geq \frac{1}{2\varepsilon^2} (\log|\mathcal{H}| + \log(\frac{2}{\delta}))$, then with probability at least $1 - \delta$ over $S \sim \mathcal{D}^m$:

$$\forall h \in \mathcal{H} : |R(h) - \hat{R}_S(h)| \leq \varepsilon \quad (13)$$

If ERM returns $h_S = \operatorname{argmin}_{h \in \mathcal{H}} \hat{R}_S(h)$ and the uniform deviation above holds with tolerance $\frac{\varepsilon}{2}$, then

$$R(h_S) \leq \min_{h \in \mathcal{H}} R(h) + \varepsilon \quad (14)$$

, which gives the corresponding excess-risk sample complexity $m \geq \frac{2}{\varepsilon^2} (\log|\mathcal{H}| + \log(\frac{2}{\delta}))$.

Proof (Guided Discovery):

Step 1: Apply Hoeffding to a single hypothesis.

Fix $h \in \mathcal{H}$. The empirical error is:

$$\hat{R}_S(h) = \frac{1}{m} \sum_{i=1}^m \mathbb{1}_{h(x_i) \neq y_i} \quad (15)$$

This is an average of i.i.d. bounded random variables in $[0, 1]$.

Pause and Think

Question: What concentration inequality should we use?

Hint: We have i.i.d. bounded variables and want to bound deviation from expectation.

Answer:

$$(91) \quad \mathbb{P} \left[\left| \frac{1}{m} \sum_{i=1}^m \ell(h; \mathbf{x}_i, \mathbf{y}_i) - \mathbb{E} \ell(h; \mathbf{x}, \mathbf{y}) \right| > \varepsilon \right] \leq 2 \exp(-2m\varepsilon^2) \quad \text{Hoeffding's inequality (Week 1) For } \varepsilon > 0:$$

Step 2: Union bound over all hypotheses.

$$\begin{aligned} \mathbb{P}_S \left[\exists h \in \mathcal{H} : |R(h) - \hat{R}_S(h)| > \varepsilon \right] &\leq \sum_{h \in \mathcal{H}} \mathbb{P} \left[|R(h) - \hat{R}_S(h)| > \varepsilon \right] \\ &\leq |\mathcal{H}| \cdot 2 \exp(-2m\varepsilon^2) \end{aligned} \quad (17)$$

Step 3: Solve for sample complexity.

Set $|\mathcal{H}| \cdot 2 \exp(-2m\varepsilon^2) \leq \delta$:

$$\log |\mathcal{H}| + \log 2 - 2m\varepsilon^2 \leq \log \delta \quad (18)$$

$$m \geq \frac{1}{2\varepsilon^2} \left(\log |\mathcal{H}| + \log \left(\frac{2}{\delta} \right) \right) \quad (19)$$

Key Takeaway

Agnostic learning requires $O(\frac{1}{\varepsilon^2})$ samples versus $O(\frac{1}{\varepsilon})$ for realizable case. This is the price of not knowing whether the target is in $\mathcal{H}$. We must ensure **uniform convergence**: all hypotheses have empirical error close to true error simultaneously.

Connection

Uniform Convergence: The bound $|R(h) - \hat{R}_S(h)| \leq \varepsilon$ holds for **all** $h \in \mathcal{H}$ simultaneously. This is stronger than pointwise convergence and is essential for ERM to work.

✓ Checkpoint

Checkpoint: Why does agnostic learning require $O(\frac{1}{\varepsilon^2})$ samples while realizable learning only needs $O(\frac{1}{\varepsilon})$? What extra challenge does the agnostic setting face?

Progress on backbone: Both realizable and agnostic bounds depend on $\log |\mathcal{H}|$. This seems to solve the problem—until we realize that real hypothesis classes (linear classifiers, neural networks, decision trees) are **infinite**. Our bounds explode to ∞ . We've hit a fundamental wall.

But this raises the central question of Week 2: **Is learning from infinite hypothesis classes impossible, or are we just measuring complexity wrong?**

Beyond Finite Hypothesis Sets

The Infinite Challenge

Thought Experiment

Consider $\mathcal{H} = \{\text{all linear classifiers in } \mathbb{R}^{100}\}$. Then $|\mathcal{H}| = \infty$, so $\log|\mathcal{H}| = \infty$. Our bounds become vacuous!

Yet intuitively, linear classifiers in $\mathbb{R}^{100}$ should be learnable—they have only 101 parameters. What's wrong?

Key insight: Counting hypotheses is too crude. We need a measure of **effective complexity** that captures how hypotheses behave on finite samples.

Example: Handwritten Digit Classification

Consider classifying 28×28 pixel images (MNIST digits 0 vs 1):

- Input space: $\mathcal{X} = \mathbb{R}^{28 \times 28} = \mathbb{R}^{784}$
- Linear classifiers: $h(x) = \text{sign}(w \cdot x + b)$ where $w \in \mathbb{R}^{784}$
- Cardinality: $|\mathcal{H}| = \infty$ (continuum of weight vectors)
- Yet linear classifiers feel “simple”—only 785 parameters

Intuition: Maybe what matters isn't **how many** hypotheses exist, but **how differently** they behave on actual data points.

Opening Challenge

Central Question: Can we replace $\log|\mathcal{H}|$ with a finite quantity that works even when $|\mathcal{H}| = \infty$?

Answer: Yes! The growth function and VC dimension provide exactly this.

Progress on backbone: The key insight is that we've been counting the wrong thing. Even though a hypothesis class might be infinite, on any **finite sample** it can only produce finitely many distinct behaviors. If we count behaviors instead of hypotheses, we might salvage learnability.

Growth Functions and Shattering

The key insight: on any finite sample of m points, even an infinite hypothesis class produces only finitely many distinct labelings. If we count behaviors instead of hypotheses, we might salvage learnability.

Dichotomies

Definition: Dichotomy

Given a set of points $S = \{x_1, \dots, x_m\} \subseteq \mathcal{X}$, a **dichotomy** (a specific labeling pattern on those points) is a labeling of these m points. The set of dichotomies realized by $\mathcal{H}$ on S is:

$$\mathcal{H}|_S = \{(h(x_1), \dots, h(x_m)) : h \in \mathcal{H}\} \subseteq \{0, 1\}^m \quad (20)$$

In words: $\mathcal{H}|_S$ is the set of all distinct label patterns (vectors of 0s and 1s) that hypotheses in $\mathcal{H}$ can produce on the points in S . Different hypotheses might produce the same labeling pattern, so $|\mathcal{H}|_S| \leq |\mathcal{H}|$.

Concrete example: If $S = \{x_1, x_2\}$ and $\mathcal{H}$ contains three hypotheses that label S as $(0, 0)$, $(0, 1)$, and $(1, 1)$, then $\mathcal{H}|_S = \{(0, 0), (0, 1), (1, 1)\}$ with $|\mathcal{H}|_S| = 3$ (even though $|\mathcal{H}| = 3$ or could be larger if other hypotheses produce the same patterns).

Key insight: We don't care about the (possibly infinite) number of hypotheses, only about the (necessarily finite) number of distinct behaviors on m points. This is the bridge from infinite to finite complexity.

Example: Linear Classifiers in $\mathbb{R}^2$

Take 3 points in general position. How many dichotomies can linear classifiers realize?

- Total possible: $2^3 = 8$
- Realizable by lines: all 8! (verify by drawing)
- Take 4 points in convex position: cannot realize XOR pattern (opposite corners positive)

Growth Function

Definition: Growth Function

The **growth function** of hypothesis class $\mathcal{H}$ is:

$$\Pi_{\mathcal{H}}(m) = \max_{x_1, \dots, x_m \in \mathcal{X}} |\mathcal{H}|_{\{x_1, \dots, x_m\}}| \quad (21)$$

Notation clarification: The outer vertical bars with $|\dots|$ denote **cardinality** (set size). The expression $\mathcal{H}|_{\{x_1, \dots, x_m\}}$ is the **restriction notation** (the set of dichotomies $\mathcal{H}$ realizes on those points, defined above).

In words: $\Pi_{\mathcal{H}}(m)$ is the maximum number of distinct labeling patterns that $\mathcal{H}$ can produce on **any** choice of m points. It measures the worst-case expressive power of $\mathcal{H}$ on m examples.

Concrete example: For threshold classifiers $\mathcal{H} = \{h_t : h_t(x) = \mathbb{1}_{x > t}\}$ on $\mathbb{R}$ and $m = 3$ points $x_1 < x_2 < x_3$, there are only 4 achievable patterns: $(0, 0, 0)$, $(1, 0, 0)$, $(1, 1, 0)$, $(1, 1, 1)$. So $\Pi_{\mathcal{H}}(3) = 4$, not $2^3 = 8$.

Properties:

- $\Pi_{\mathcal{H}}(m) \leq 2^m$ (at most all possible labelings exist)
- $\Pi_{\mathcal{H}}(m)$ is non-decreasing in m (more points allow at least as many patterns)
- **Distribution-independent:** purely combinatorial, doesn't depend on $\mathcal{D}$

Definition: Shattering

A set $S = \{x_1, \dots, x_m\}$ is **shattered** by $\mathcal{H}$ if $|\mathcal{H}|_S| = 2^m$ —that is, $\mathcal{H}$ realizes **all** possible dichotomies (all 2^m labeling patterns) on S .

In words: $\mathcal{H}$ shatters S if for every possible way to label the m points in S with 0s and 1s, there exists at least one hypothesis $h \in \mathcal{H}$ that produces exactly that labeling.

Concrete example: Three non-collinear points in $\mathbb{R}^2$ are shattered by linear classifiers (lines can separate them in all 8 possible ways). Four points in convex position are **not** shattered (no line can achieve the XOR pattern where opposite corners are labeled the same).

 Construct Your Own**Compute growth functions by hand:**

1. **Positive rays** on $\mathbb{R}$: $\mathcal{H} = \{h_t : h_t(x) = \mathbb{1}_{x > t}, t \in \mathbb{R}\}$
 - For m points $x_1 < x_2 < \dots < x_m$, how many dichotomies?
 - Hint: Each hypothesis creates a “cut” at some threshold
 - Answer: $m + 1$ (why not 2^m ?)
2. **Two thresholds:** $\mathcal{H} = \{h_{a,b} : h(x) = \mathbb{1}_{a < x < b}, a \leq b\}$
 - For 3 points $x_1 < x_2 < x_3$, list all achievable dichotomies
 - Which labelings are impossible? (e.g., can we get $(1, 0, 1)$?)
3. **Verify:** Does your growth function match 2^m for small m ? When does it drop below?

Pause and Think

Question: If $\mathcal{H}$ shatters some set of size m , what can we say about $\Pi_{\mathcal{H}}(m)$?

Hint: The growth function is the maximum over all sets of size m .

Answer:

We have $\Pi_{\mathcal{H}}(m) = 2^m$ (the maximum is achieved).

✓ Checkpoint

Checkpoint: Explain the difference between counting hypotheses ($|\mathcal{H}|$) and counting dichotomies ($\Pi_{\mathcal{H}}(m)$). Why is the latter more useful for infinite classes?

Progress on backbone: The growth function counts behaviors on samples of size m . But we need **one number**—like $|\mathcal{H}|$ for finite classes—that summarizes a hypothesis class's entire complexity profile. This brings us to VC dimension: the critical threshold where growth transitions from exponential to polynomial.

VC Dimension

VC dimension (Vapnik-Chervonenkis dimension, named after the mathematicians who introduced it) asks: what's the largest set that our hypothesis class can label in all possible ways? This single number will replace $|\mathcal{H}|$ in our sample complexity bounds and completely characterize learnability.

Definition and Intuition

Definition: VC Dimension

The **Vapnik-Chervonenkis (VC) dimension** of hypothesis class $\mathcal{H}$ is:

$$\text{VCdim}(\mathcal{H}) = \max\{m : \Pi_{\mathcal{H}}(m) = 2^m\} \quad (22)$$

Equivalently: the size of the largest set that can be shattered by $\mathcal{H}$ (where **shattered** means $\mathcal{H}$ can realize all 2^m possible labelings of the m points).

In words: VC dimension is the largest number d such that there exists at least one set of d points that $\mathcal{H}$ can label in all 2^d possible ways. For all sets of size $d + 1$, at least one labeling is impossible.

Critical distinction—common source of confusion:

- If $\text{VCdim}(\mathcal{H}) = d$, then there **exists** at least one set of size d that is shattered
- It is **not** required that all sets of size d be shattered (most may not be!)
- **No** set of size $d + 1$ can be shattered (this is a universal statement)

Concrete example: Linear classifiers in $\mathbb{R}^2$ have $\text{VCdim} = 3$. This means: (1) there exists a set of 3 points (e.g., vertices of a triangle) that can be separated in all 8 ways, (2) not every set of 3 points can be shattered (3 collinear points cannot), (3) no set of 4 points can be fully shattered.

Wrong Path: Common Misconception

“VC dimension equals the number of parameters.”

Counterexample: $\mathcal{H} = \{\text{sign}(\sin(\omega x)) : \omega \in \mathbb{R}\}$ has 1 parameter but $\text{VCdim}(\mathcal{H}) = \infty$! High-frequency oscillations can realize arbitrary labels on carefully chosen finite point sets.

Lesson: VC dimension measures geometric complexity, not parametric complexity.

Computing VC Dimension: Strategy

To prove $\text{VCdim}(\mathcal{H}) = d$, you must establish two bounds:

1. **Lower bound** ($\text{VCdim} \geq d$): Exhibit one specific set of size d that is shattered (constructive —show an example)

2. **Upper bound** ($\text{VCdim} < d + 1$): Prove no set of size $d + 1$ can be shattered (proof by contradiction or structural argument—show impossibility for **all** possible sets)

Key Takeaway

The lower bound is constructive (show one example configuration that works). The upper bound requires a general argument (prove impossibility for **all** configurations of $d + 1$ points, regardless of placement).

Mental Model: VC Dimension Calculation Strategy

Decision tree for computing VC dimension:

1. **Start with intuition:** Count degrees of freedom (parameters) as initial guess
2. **Lower bound:** Construct specific point configuration that can be shattered
 - Use geometric intuition (corners, extremes, symmetry)
 - Verify all 2^d labelings are achievable
3. **Upper bound:** Find structural constraint that prevents shattering
 - Connectivity arguments (intervals)
 - Dimension arguments (Radon's theorem for hyperplanes)
 - Extremal point arguments (rectangles)
4. **Verify:** Check boundary cases and edge conditions

Common patterns:

- Linear separators in $\mathbb{R}^d$: VC dim = $d + 1$ (use affine independence)
- Unions of k regions: VC dim grows with k (construct separate shatterable sets)
- Constrained shapes: VC dim limited by structural properties (connectivity, convexity)

Example 1: Intervals on $\mathbb{R}$

Example: Intervals = Biomarker Thresholds

Recall disease diagnosis: $\mathcal{H} = \{h_t : h_t(x) = \mathbb{1}_{x>t}\}$ classifies based on whether biomarker exceeds threshold t .

This is equivalent to $\mathcal{H} = \{[a, b] : a \leq b, a, b \in \mathbb{R}\}$ (predict positive inside interval, negative outside).

Let $\mathcal{H} = \{[a, b] : a \leq b, a, b \in \mathbb{R}\}$ be axis-aligned intervals on the real line.

Lower bound ($d \geq 2$): Take $x_1 < x_2$.

- $(0, 0)$: $[x_2 + 1, x_2 + 2]$ ✓
- $(1, 0)$: $[x_1, x_1 + \frac{x_2 - x_1}{2}]$ ✓
- $(0, 1)$: $[x_1 + \frac{x_2 - x_1}{2}, x_2]$ ✓
- $(1, 1)$: $[x_1, x_2]$ ✓

All 4 dichotomies realized!

Upper bound ($d < 3$): Take any $x_1 < x_2 < x_3$.

Can we realize $(1, 0, 1)$ (label endpoints positive, middle negative)?

No! An interval is connected, so if x_1 and x_3 are in the interval, then x_2 must also be in it.

Conclusion: $\text{VCdim}(\text{intervals}) = 2$

Student Challenge

Exercise: Compute $\text{VCdim}(\text{unions of } k \text{ intervals})$. Hint: try $k = 2$ first.

Example 2: Linear Classifiers in $\mathbb{R}^d$ **Example: Hyperplanes**

Let $\mathcal{H} = \{\text{sign}(w \cdot x + b) : w \in \mathbb{R}^d, b \in \mathbb{R}\}$ be linear classifiers in $\mathbb{R}^d$.

Lower bound ($\text{VCdim} \geq d + 1$): Choose any $d + 1$ affinely independent points in $\mathbb{R}^d$, for example the origin and the d standard basis vectors.

For any labeling of these points, the convex hull of the positive points is disjoint from the convex hull of the negative points because affine independence makes every point a vertex of the simplex. The hyperplane separation theorem gives an affine hyperplane separating the two hulls, so that labeling is realized by some $\text{sign}(w \cdot x + b)$.

Therefore all 2^{d+1} dichotomies on this set are realizable.

Upper bound ($d < d + 2$): Use Radon's theorem.

Productive Struggle

Before reading Radon's theorem proof: Try to see why $d + 2$ points in $\mathbb{R}^d$ must have “too much overlap.”

Intuition builder:

- In $\mathbb{R}^1$: Take 3 points. Can you always partition them so their “spans” overlap?
- In $\mathbb{R}^2$: Take 4 points. Can you always find a partition where convex hulls intersect?
- General $\mathbb{R}^d$: Why does $d + 2$ points exceed some “capacity”?

Key insight: This is about **linear dependence**. We have $d + 2$ points living in a d -dimensional space. The augmented vectors $(x_i, 1) \in \mathbb{R}^{d+1}$ must be linearly dependent (why?). How does this force an intersection?

Try to sketch the argument using linear algebra before reading the formal proof.

Theorem: Radon's Theorem

Any $d + 2$ points in $\mathbb{R}^d$ can be partitioned into two disjoint sets X_1, X_2 whose **convex hulls** (the smallest convex sets containing each point set) intersect.

In words: Given too many points ($d + 2$) in a d -dimensional space, you can always split them into two groups such that their convex hulls overlap—meaning no hyperplane can perfectly separate these two groups.

Proof sketch: Augment each point $x_i \in \mathbb{R}^d$ to $(x_i, 1) \in \mathbb{R}^{d+1}$. Since we have $d + 2$ vectors in a $(d + 1)$ -dimensional space, they must be **linearly dependent** (a set of $d + 2$ vectors in $\mathbb{R}^{d+1}$ cannot all be independent). Therefore, there exist coefficients α_i , not all zero, with:

$$\sum_{i=1}^{d+2} \alpha_i (x_i, 1) = (0, 0) \quad (23)$$

This means $\sum_i \alpha_i x_i = 0$ and $\sum_i \alpha_i = 0$ (from the last component). Separate the positive coefficients from negative: let $I_+ = \{i : \alpha_i > 0\}$ and $I_- = \{i : \alpha_i < 0\}$. Normalize each side to get convex combinations that represent the same point, proving the hulls intersect.

Applying Radon to VC dimension: If $\text{conv}(X_1) \cap \text{conv}(X_2) \neq \emptyset$ (the convex hulls intersect at some point p), then no hyperplane can separate X_1 from X_2 —any hyperplane must have points from both sets on at least one side (since p belongs to both hulls). Thus, we cannot realize the dichotomy that labels all of X_1 positive and all of X_2 negative.

Conclusion: $\text{VCdim}(\text{hyperplanes in } \mathbb{R}^d) = d + 1$

Connection

Geometric intuition: A hyperplane in $\mathbb{R}^d$ has $d + 1$ degrees of freedom (d for normal vector, 1 for offset). VC dimension matches this, but the connection is not always so direct!

Example 3: Axis-Aligned Rectangles in $\mathbb{R}^2$

Example: Rectangles

Let $\mathcal{H} = \{[a_1, b_1] \times [a_2, b_2] : a_i \leq b_i\}$ be axis-aligned rectangles in the plane.

Lower bound ($d \geq 4$): Take 4 points in diamond configuration:

- $(0, 1)$ (top)
- $(1, 0)$ (right)
- $(0, -1)$ (bottom)
- $(-1, 0)$ (left)

All $2^4 = 16$ dichotomies are realizable (verify by drawing rectangles).

Upper bound ($d < 5$): Take any 5 points. Choose a leftmost, rightmost, topmost, and bottommost point (ties can be broken arbitrarily). These at most 4 extreme points determine a minimal axis-aligned rectangle containing all 5 points.

Since there are 5 points, at least one point is not among the chosen extremes. Label the chosen extreme points positive and this remaining point negative.

Any axis-aligned rectangle containing the extreme points must contain the entire bounding rectangle, hence must also contain the negative point.

Conclusion: $\text{VCdim}(\text{rectangles}) = 4$

Student Challenge

Exercise: What is $\text{VCdim}(\text{convex polygons in } \mathbb{R}^2)$? Hint: it's infinite!

 Construct Your Own

Design your own hypothesis class:

1. **Target VC dimension 3:** Create a hypothesis class with exactly $\text{VCdim}(\mathcal{H}) = 3$
 - Hint: Think about unions of intervals on $\mathbb{R}$
 - Prove both bounds: show 3 points can be shattered, show 4 cannot
2. **Create counterexample:** Find a hypothesis class where the number of parameters does not equal VC dimension
 - Example: $\mathcal{H} = \{\text{sign}(\sin(\omega x)) : \omega \in [0, M]\}$ for large M
 - How does VC dimension grow with M ?
3. **Geometric construction:** Design a point configuration in $\mathbb{R}^2$ that:
 - Can be shattered by circles (center + radius)
 - Cannot be shattered by axis-aligned rectangles

 Confidence Check

Rate your understanding (1-5):

- ☐ I can compute VC dimension for new concept classes
- ☐ I understand why VC dimension is not always equal to the number of parameters
- ☐ I can prove both lower and upper bounds on VC dimension

Progress on backbone: We can now compute VC dimension for specific hypothesis classes. But a crucial question remains: **why does finite VC dimension guarantee learnability?** We need to connect VC dimension back to sample complexity. Sauer's lemma provides this mathematical bridge.

Sauer's Lemma

Sauer's lemma is the theoretical centerpiece: it proves that finite VC dimension implies polynomial growth. This converts our intuition—"behavior-counting should work"—into rigorous mathematics.

Productive Struggle

Before diving into Sauer's lemma: Pause and try to formalize your intuition.

We know:

- $\text{VCdim}(\mathcal{H}) = d$ means no set of $d + 1$ points can be shattered
- Growth function $\Pi_{\mathcal{H}}(m)$ counts maximum dichotomies on m points
- We want to show finite VC dimension implies polynomial growth

Challenge: How would you connect these concepts?

Try to sketch why "cannot shatter $d + 1$ points" should lead to " $\Pi_{\mathcal{H}}(m)$ grows polynomially." What structural property prevents exponential growth?

Hint: Think about what happens when you add one more point to a set. If you can't shatter any $(d + 1)$ -subset, how does this constrain the number of new dichotomies?

Spend 5-10 minutes thinking before reading the proof. The struggle will make the elegant proof structure more memorable.

The Fundamental Dichotomy

Recall from Section 6 that the growth function is defined as:

$$\Pi_{\mathcal{H}}(m) = \max_{S \subseteq \mathcal{X}, |S|=m} |\mathcal{H}|_S| \quad (24)$$

That is, $\Pi_{\mathcal{H}}(m)$ is the **maximum** number of distinct labelings $\mathcal{H}$ can realize on **any** set of m points. Sauer's lemma bounds this maximum:

Theorem: Sauer-Shelah Lemma

Let $\mathcal{H}$ have $\text{VCdim}(\mathcal{H}) = d < \infty$. Then for all $m \in \mathbb{N}$:

$$\Pi_{\mathcal{H}}(m) = \max_{S \subseteq \mathcal{X}, |S|=m} |\mathcal{H}|_S| \leq \sum_{i=0}^d \binom{m}{i}. \quad (25)$$

For $1 \leq d \leq m$, this is at most $\left(\frac{em}{d}\right)^d$; for $d = 0$, the sum is 1.

Significance: Two regimes only!

- If $\text{VCdim}(\mathcal{H}) = d < \infty$: $\Pi_{\mathcal{H}}(m) = O(m^d)$ (polynomial)
- If $\text{VCdim}(\mathcal{H}) = \infty$: $\Pi_{\mathcal{H}}(m) = 2^m$ (exponential)

Key Takeaway

Sauer's lemma shows that VC dimension completely characterizes the asymptotic behavior of the growth function. Finite VC dimension implies polynomial growth, which is the key to learnability.

Example: Why Polynomial Growth Enables Learning

Credit scoring with $d = 10$ features (income, credit history, etc.):

- Hypothesis class: linear classifiers in $\mathbb{R}^{10}$
- VC dimension: $d = 11$
- Growth on $m = 1000$ samples: $\Pi_{\mathcal{H}}(1000) \leq \left(\frac{e \cdot 1000}{11}\right)^{11} \approx 10^{24}$
- Compare to exponential: $2^{1000} \approx 10^{301}$

Polynomial growth means we can apply union bounds without the probability exploding. The gap between 10^{24} and 10^{301} is the difference between learnable and unlearnable!

Proof Sketch of Sauer's Lemma

Proof idea (by induction on $m + d$):

Base cases: If $m = 1$: $\Pi_{\mathcal{H}}(1) \leq 2$. If $d = 0$: $\Pi_{\mathcal{H}}(m) \leq 1$.

Inductive step: Fix $S = \{x_1, \dots, x_m\}$ and let $S' = \{x_1, \dots, x_{m-1}\}$. Project each dichotomy on S down to S' by forgetting x_m 's label:

1. **Unique projections:** Labeling on S' has only one extension to x_m (bounded by $\Pi_{\mathcal{H}}(m - 1)$)
2. **Duplicate projections:** Labeling on S' has both extensions. Let B be the set of such duplicates. If B shattered a subset of S' of size d , adding x_m would shatter size $d + 1$ (contradiction!). So $\text{VCdim}(B) \leq d - 1$.

This gives the recurrence $\Pi_m(d) \leq \Pi_{m-1}(d) + \Pi_{m-1}(d - 1)$. Applying induction and Pascal's identity yields the result.

The second inequality $\sum_{i=0}^d \binom{m}{i} \leq \left(\frac{em}{d}\right)^d$ for $1 \leq d \leq m$ follows from standard binomial bounds. ■

Connection

Proof technique: This is a beautiful example of **double induction** (on $m + d$) combined with a **projection argument** (removing one point and analyzing the structure).

✓ Checkpoint

Checkpoint: State Sauer's lemma in one sentence and explain why polynomial growth (rather than exponential) makes infinite hypothesis classes learnable.

Progress on backbone: Sauer's lemma proved that finite VC dimension implies polynomial growth: $\Pi_{\mathcal{H}}(m) \leq \sum_{i=0}^d \binom{m}{i} = O(m^d)$. Now we complete the circle: plug this bound into our PAC framework to get sample complexity guarantees for infinite hypothesis classes.

VC Dimension and Generalization

The payoff: replace $\log|\mathcal{H}|$ with d in our bounds. Infinite hypothesis classes are learnable if—and only if—their VC dimension is finite.

The Fundamental Theorem

Theorem: Fundamental Theorem of PAC Learning (Informal Statement)

A hypothesis class $\mathcal{H}$ (for binary classification) is PAC-learnable if and only if $\text{VCdim}(\mathcal{H}) < \infty$.

Moreover, the sample complexity is $\Theta\left(\frac{d + \log(\frac{1}{\delta})}{\epsilon^2}\right)$ in the agnostic case and $\Theta\left(\frac{d + \log(\frac{1}{\delta})}{\epsilon}\right)$ in the realizable case, where $d = \text{VCdim}(\mathcal{H})$.

This week's focus: We prove the upper bound (finite VC dimension implies learnability) via the generalization bound below. The converse (infinite VC dimension implies unlearnability) and the precise characterization are developed in Week 3.

→ Week 3 Preview: Fundamental Theorem

Week 3 presents the complete Fundamental Theorem including the necessity direction (why infinite VC dimension prevents learning) and the connection to uniform convergence.

Generalization Bound via VC Dimension

Theorem: VC Generalization Bound (Simplified)

Let $\mathcal{H}$ have $\text{VCdim}(\mathcal{H}) = d < \infty$ with $1 \leq d \leq m$. For any $\delta > 0$ and any distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$:

With probability at least $1 - \delta$ over $S \sim \mathcal{D}^m$:

$$\forall h \in \mathcal{H} : R(h) \leq \hat{R}_S(h) + \sqrt{\frac{8d \log\left(\frac{em}{d}\right) + 8 \log\left(\frac{4}{\delta}\right)}{m}} \quad (26)$$

For edge cases such as $d = 0$ or $d > m$, it is safer to use the growth-function version of the bound. The constants above are one common simplified form, not a canonical constant-optimal statement.

 Theorem Landscape

How the three main bounds relate:

Setting	Sample Complexity	Key Assumption	Tightness
Realizable Finite	$m = O\left(\frac{\log \mathcal{H} + \log(\frac{1}{\delta})}{\varepsilon}\right)$	Perfect h^* exists	$\Theta\left(\frac{1}{\varepsilon}\right)$
Agnostic Finite	$m = O\left(\frac{\log \mathcal{H} + \log(\frac{1}{\delta})}{\varepsilon^2}\right)$	None (worst case)	$\Theta\left(\frac{1}{\varepsilon^2}\right)$
VC Dimension	$m = O\left(\frac{d \log(\frac{m}{d}) + \log(\frac{1}{\delta})}{\varepsilon^2}\right)$	Finite VC dim	$\Theta\left(\frac{d}{\varepsilon^2}\right)$

Conceptual flow:

1. **Realizable** $\rightarrow$ **Agnostic**: Removing realizability assumption costs factor of ε (need $\frac{1}{\varepsilon^2}$ vs $\frac{1}{\varepsilon}$)
2. **Finite** $\rightarrow$ **Infinite**: Replace $\log|\mathcal{H}|$ with $d \log(\frac{m}{d})$ via Sauer's lemma
3. **All bounds**: Share union bound structure—probability of bad event $\leq$ (complexity measure) $\times$ (single-hypothesis failure probability)

Which bound to use?

- If $|\mathcal{H}|$ small and realizable known: use realizable bound
- If $|\mathcal{H}|$ small but noisy labels: use agnostic bound
- If $|\mathcal{H}| = \infty$ but structured: use VC bound
- If data-dependent tightness needed: use Rademacher (Week 3)

Proof sketch:

1. Replace $\log|\mathcal{H}|$ in the finite case with $\log \Pi_{\mathcal{H}}(2m)$
2. Use Sauer's lemma: $\log \Pi_{\mathcal{H}}(2m) \leq d \log\left(\frac{2em}{d}\right)$
3. Apply symmetrization and uniform convergence arguments
4. Simplify to get the stated bound

$\rightarrow$ **Week 3 Preview: Symmetrization**

Symmetrization technique (detailed in Week 3): To bound the deviation $|R(h) - \hat{R}_S(h)|$, we introduce a “ghost sample” S' drawn independently from the same distribution. The key insight: $|R(h) - \hat{R}_S(h)|$ can be bounded by $|\hat{R}_S(h) - \hat{R}_{S'}(h)|$, the difference between two empirical estimates. This replaces an unknown expectation with a computable empirical quantity, enabling tighter analysis.

The factor of $2m$ in step 1 comes from analyzing both the actual sample S and the ghost sample S' together.

🏗️ Scaffolded Derivation (3 steps)

Deriving sample complexity from VC bound:

Step 1: Start with generalization bound

$$R(h) \leq \hat{R}_S(h) + \sqrt{\frac{8d \log\left(\frac{em}{d}\right) + 8 \log\left(\frac{4}{\delta}\right)}{m}} \quad (27)$$

Step 2: Set right-hand side $\leq \varepsilon$ for ERM

- If we run ERM: $\hat{R}_S(h_S) \leq \min_{h \in \mathcal{H}} \hat{R}_S(h) \leq \min_{h \in \mathcal{H}} R(h)$
- Want: $R(h_S) \leq \min_{h \in \mathcal{H}} R(h) + \varepsilon$
- Need: $\sqrt{\frac{8d \log\left(\frac{em}{d}\right) + 8 \log\left(\frac{4}{\delta}\right)}{m}} \leq \frac{\varepsilon}{2}$

Step 3: Solve for m

$$\begin{aligned} 8d \log\left(\frac{em}{d}\right) + 8 \log\left(\frac{4}{\delta}\right) &\leq \left(\frac{\varepsilon}{2}\right)^2 m \\ d \log\left(\frac{m}{d}\right) + \log\left(\frac{1}{\delta}\right) &\leq O(\varepsilon^2 m) \\ m &\geq O\left(\frac{d \log\left(\frac{m}{d}\right) + \log\left(\frac{1}{\delta}\right)}{\varepsilon^2}\right) \end{aligned} \quad (28)$$

For large m , the $\log\left(\frac{m}{d}\right)$ term is much smaller than m , giving the final form:

$$m = O\left(\frac{d + \log\left(\frac{1}{\delta}\right)}{\varepsilon^2}\right) \quad (29)$$

Key Takeaway

Sample complexity for VC dimension d :

$$m = O\left(\frac{d \log\left(\frac{1}{\varepsilon}\right) + \log\left(\frac{1}{\delta}\right)}{\varepsilon^2}\right) \quad (30)$$

The key term is $\frac{d}{\varepsilon^2}$, not $|\mathcal{H}|$. This makes infinite hypothesis sets learnable!

Comparison of Bounds

Example: Sample Complexity Comparison

For $\varepsilon = 0.1$, $\delta = 0.05$:

- Finite $\mathcal{H}$ with $|\mathcal{H}| = 2^{100}$: $m \approx 2000$ (realizable) or $m \approx 20000$ (agnostic)
- Linear classifiers in $\mathbb{R}^{100}$: $d = 101$, so $m \approx 10000$
- Credit scoring with $d = 10$ features: $m \approx 1000$
- Single biomarker threshold (intervals): $d = 2$, so $m \approx 200$

Trade-off: Complex models (high d) need more data but can capture richer patterns. Simple models (low d) learn from less data but may underfit.

Key insight: VC dimension provides much tighter bounds for structured hypothesis classes than naive cardinality counting!

Lower Bounds**Theorem: Lower Bounds for PAC Learning**

Let $\mathcal{H}$ be a hypothesis class with $\text{VCdim}(\mathcal{H}) = d \geq 1$.

Realizable case: Any learning algorithm for $\mathcal{H}$ requires sample complexity at least

$$m \geq \Omega\left(\frac{d + \log(\frac{1}{\delta})}{\varepsilon}\right) \quad (31)$$

to achieve error ε with probability $1 - \delta$.

Agnostic case: Any learning algorithm for $\mathcal{H}$ requires sample complexity at least

$$m \geq \Omega\left(\frac{d + \log(\frac{1}{\delta})}{\varepsilon^2}\right) \quad (32)$$

to achieve excess error ε with probability $1 - \delta$.

Implication: The VC dimension upper bounds (from previous theorem) are tight up to logarithmic factors.

Implication: The VC upper bounds are tight up to logarithmic factors in their respective regimes.

Proof technique: Probabilistic method (Erdős). Construct a distribution on a shattered set and show that, with too few samples, any algorithm cannot reliably distinguish between many possible labelings.

Key Takeaway

The VC dimension bounds are **optimal**. We cannot do better than $\Theta(\frac{d}{\varepsilon^2})$ samples in the agnostic case or $\Theta(\frac{d}{\varepsilon})$ in the realizable case.

We've completed the journey: from finite hypothesis counting to VC dimension's behavior-counting, proving that complexity—not cardinality—determines learnability.

Summary and Connections

Key Insights

1. **PAC framework** formalizes “when is learning possible?” with polynomial sample complexity
2. **Finite hypothesis sets** are learnable with $O(\log|\mathcal{H}|)$ samples
3. **VC dimension** extends learnability to infinite hypothesis sets
4. **Sauer’s lemma** shows finite VC dimension implies polynomial growth
5. **Generalization bounds** scale with $\frac{d}{m}$, not $|\mathcal{H}|$

Connection

To Week 3: VC dimension is distribution-independent. Next week we introduce **Rademacher complexity**, a data-dependent measure that can give tighter bounds for specific distributions.

To Week 4: VC dimension motivates **structural risk minimization**: balance empirical error with model complexity.

To Later Weeks: VC dimension of neural networks (Week 10-11) and ensemble methods (Week 6-7).

Pause and Think

Question: Trace the backbone: How did we go from “infinite $\mathcal{H}$ means $\log|\mathcal{H}| = \infty$ ” to “infinite $\mathcal{H}$ is learnable”?

Hint: The key was changing what we count.

Answer:

We stopped counting hypotheses (cardinality) and started counting behaviors on samples (growth function). Sauer’s lemma proved finite VC dimension implies polynomial growth, which makes union bounds work. The lesson: effective complexity, not size, determines learnability.

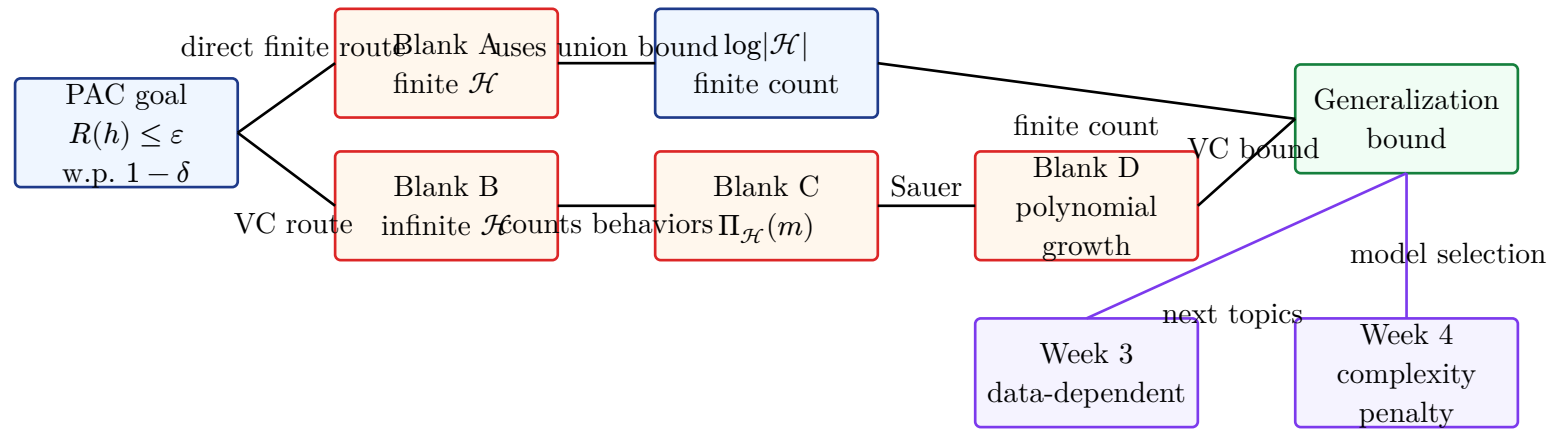
Synthesis Challenge

Design a hypothesis class with VC dimension exactly 5. Prove both bounds: (1) exhibit a specific set of 5 points that can be shattered, and (2) prove no set of 6 points can be shattered.

Concept Map Exercise

Fill in the missing labels in the diagram. The red nodes labeled **Blank A–D** are placeholders for concepts/theorems you should identify. Read the edges as a directed conceptual flow from left to right: each edge explains how one idea leads to the next bound or complexity measure.

Week 2 Concept Map Exercise



Pause and Think

Question: Complete Blanks A–D and explain the contrast between the two branches. Are the arrows directed? Should every edge have a label?

Hint: Blanks A–D are missing concept labels: A and B name the two routes; C and D name the combinatorial quantities/theorems used in the infinite-class route. The edges are intended as directed left-to-right dependencies, not undirected associations. Edge labels are short hints showing what role each dependency plays.

Answer:

A = finite-hypothesis union bound: replace complexity by $\log|\mathcal{H}|$. B = VC dimension: a combinatorial size measure for infinite classes. C = growth function: number of label patterns on m points. D = Sauer's lemma: finite VC dimension d implies $\Pi_{\mathcal{H}}(m) \leq \left(\frac{em}{d}\right)^d$. Contrast: finite PAC bounds depend on the literal number of hypotheses, while VC bounds depend on the number of distinct behaviors on samples. This is why infinite classes can still generalize.

Closing Challenge

Opening Challenge

Revisiting the Opening Question:

You trained a neural network (10,000 neurons) on 1,000 images with 99% training accuracy. Should you deploy it in production?

VC analysis suggests caution:

- Neural networks have very high VC dimension (often proportional to number of weights)
- 10,000 neurons $\approx$ millions of parameters
- VC dimension could be in the millions
- Sample complexity $\approx \frac{d}{\epsilon^2} \approx$ millions of samples needed!
- Yet you only have 1,000 samples...

Paradox: Deep learning works in practice despite violating classical VC bounds.

Resolution preview:

- Week 3 (Rademacher complexity): data-dependent bounds can be much tighter than worst-case VC bounds
- Week 4 (Structural risk minimization): model selection balances complexity and fit
- Week 11 (Deep learning theory): implicit regularization, compression bounds, and why VC dimension is too pessimistic for neural networks

Deployment decision: Use held-out validation sets, check for distribution shift, measure prediction uncertainty, and monitor performance in production. Theory guides intuition but doesn't replace empirical validation.

→ Week 3 Preview: Rademacher Complexity

Next week introduces Rademacher complexity, a data-dependent complexity measure that can give much tighter generalization bounds than VC dimension for specific distributions and datasets.

→ Week 11 Preview: Deep Learning Theory

Week 11 resolves the paradox: modern deep learning theory explains why neural networks generalize despite having astronomical VC dimension, through implicit regularization, norm-based bounds, and compression arguments.